Python
Iterative Statements

Objective

- range() function
- Loops
- while loop
- for loop
- Nested loop
- Loop Control Statements
  - Break
  - Continue
  - Pass

range() function

- The range() function is used to generate the sequence of the numbers. If we pass the range(10), it will generate the numbers from 0 to 9.

Syntax:range(start,stop,step size)

  - The start represents the beginning of the iteration. default value is 0.
  - The stop represents that the loop will iterate till stop-1. The range(1,5) will generate numbers 1 to 4 iterations. It is optional.
  - The step size is used to skip the specific numbers from the iteration. It is optional to use. By default, the step size is 1. It is optional.

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Programs to print numbers in sequence.

Example-1:

```
for i in range(10):
    print(i,end = ' ')
```

Output:

0 1 2 3 4 5 6 7 8 9

Example:-2:

```
for i in range(3,10,2):
        print(i)
```

Output:

3
5
7
9

Loops

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Advantages of loops

There are the following advantages of loops in Python.

- It provides code re-usability.
- Using loops, we do not need to write the same code again and again.
- Using loops, we can traverse over the elements of data structures (array or linked lists).

Python has two primitive loop commands:

- while loops
- for loops
- nested loops

while loop

With the while loop we can execute a set of statements as long as a condition is true.

Syntax:

```
while expression:
   statements
```

Example:

```
i=1
#The while loop will iterate until condition becomes false.
while(i<=3):
   print(i)
   i=i+1
```

Output:

```
1
2
3
```


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Infinite while loop

If the condition is given in the while loop never becomes false, then the while loop will never terminate, and it turns into the infinite while loop.

Any non-zero value in the while loop indicates an always-true condition.

Example 1

```
while (1):
    print("Hi! we are inside the infinite while loop")
```

Output:

```
Hi! we are inside the infinite while loop
Hi! we are inside the infinite while loop
...
Infinite Time
```

Note:

ctrl+c is used to come out of infinite for loop

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Using else with while loop

- Python allows us to use the else statement with the while loop.
- The else block is executed when the condition given in the while statement becomes false.
- if the while loop is broken using break statement, then the else block will not be executed, and the statement present after else block will be executed.
- The else statement is optional to use with the while loop.

Example :

```
i=1
while(i<=3):
    print(i)
    i=i+1
else:
    print("The while loop exhausted")
```

Output:

```
1
2
3
The while loop exhausted
```

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Example:

```
i=1
while(i<=5):
   print(i)
   i=i+1
   if i==3:
      break;
else:
   print("The while loop exhausted")
```

Output:

```
1
2
```

for loop

The for loop in Python is used to iterate the statements or a part of the program several times.

It is frequently used to traverse the data structures like list, tuple, or dictionary.

Syntax :

```
for iterating_var in sequence:
   statement(s)
```

Example-1:

```
str = "Python"
for i in str:
   print(i)
```

Output:

```
P
y
t
h
o
n
```


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Example 2:
Program to print table of given number.

```
n = int(input("Enter the number "))
for i in range(1,11):
    c = n*i
    print(n,"*",i,"=",c)
```

Output:

```
Enter the number 10
10 * 1 = 10
10 * 2 = 20
10 * 3 = 30
10 * 4 = 40
10 * 5 = 50
10 * 6 = 60
10 * 7 = 70
10 * 8 = 80
10 * 9 = 90
10 * 10 = 100
```

Using else statement with for loop

- We can write else statement with the for loop which can be executed only when all the iterations are exhausted.
- If loop contains any of the break statement then the else statement will not be executed.

Example:

```
for i in range(0,5):
   print(i)
else:
   print("for loop completely exhausted, since there is no break.")
```

Output:

```
0
1
2
3
4
for loop completely exhausted, since there is no break.
```

The for loop completely exhausted, since there is no break.

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Example 2

```
for i in range(0,5):
   print(i)
   break;
else:
    print("for loop is exhausted");
print("The loop is broken due to break statement...came out of the loop")
```

Output:

0

The loop is broken due to break statement...came out of the loop

In the above example, the loop is broken due to the break statement; therefore, the else statement will not be executed. The statement present immediate next to else block will be executed.

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Nested loop

- Python allows us to nest any number of for loops inside a for loop.
- The inner loop is executed n number of times for every iteration of the outer loop.

Syntax

```
for iterating_var1 in sequence:  #outer loop
   for iterating_var2 in sequence:  #inner loop
      #block of statements
#Other statements
```


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Nested for loop

Example- 1:

Nested for loop

```
# User input for number of rows
rows = int(input("Enter the rows:"))
# Outer loop will print number of rows
for i in range(0,rows+1):
# Inner loop will print number of Astrisk
    for j in range(i):
        print("*",end = '')
    print()
```

Output:

```
Enter the rows:5
*
**
***
****
*****
```


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Nested while loop

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The syntax for a nested while loop :

```
while expression:
   while expression:
      statement(s)
      statement(s)
```

Example

```
i=1
while(i<3):
   j=1
   while(j<3):
      print(i,j,sep="-")
      j=j+1
   i=i+1
```

Output:

```
1-1
1-2
2-1
2-2
```


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Loop Control Statements

- Break
- Continue
- Pass

Break

break statement

Terminates the loop statement and transfers execution to the statement immediately following the loop.

Example:

```
for char in 'Python':
    if (char == 'y'):
        break
    print('Current character: ', char)
```

output:

```
Current character:  P
```

here loop break at y and then it skip all iterations.

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Continue

continue statement

Causes the loop to skip the current iteration after continue and the continue with remaining iterations.

Example:

```
for char in 'Python':
    if (char == 'y'):
        continue
    print('Current character: ', char)
```

output:

```
Current character:  P
Current character:  t
Current character:  h
Current character:  o
Current character:  n
```

Here y is skipped.

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Pass

pass statement

The pass statement in Python is used when a statement is required syntactically but you do not want any command or code to execute.

Example:

```
for char in 'Python':
    if (char == 'h'):
        pass
    print('Current character: ', char)
```

output:

```
Current character:  P
Current character:  y
Current character:  t
Current character:  h
Current character:  o
Current character:  n
```


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Summary

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- Loops
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Thank You